Certified thresholds for positive quantum capacity of the depolarizing, independent X–Z and two-Pauli channels

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A qubit Pauli channel has positive quantum capacity Q if some input to n uses of the channel has positive coherent information, and a rigorous evaluation of the coherent information of one explicit input proves it. For three one-parameter families we certify such inputs in interval arithmetic and obtain lower bounds on the threshold $p_{\rm th} = \sup\{p : Q > 0\}$. For the depolarizing channel, $Q > 0$ for every $p \le 0.06688201$ per Pauli error (total error $3p = 0.20064603$), with an input on $n = 480$ qubits. For the channel with independent bit and phase flips (independent X–Z, $p \le 1/2$), $Q > 0$ for every $p \le 0.12770564$ ($n = 320$). For the two-Pauli channel, $Q > 0$ at $p = 0.12677472$ ($n = 320$). The largest depolarizing value certified before was $p = 0.064956$, at $n = 45$. Ahmed *et al.* claim a larger depolarizing bound, $p = 0.0676387$ ($3p = 0.2029161$): they prove that many-copy inputs converge to codes for a Gaussian channel and evaluate one Gaussian coherent information, about 10^{-12}, numerically, without an explicit n or error bounds for the numerical step. Our depolarizing value lies below theirs. It is the largest for which positivity is certified at an explicit finite n, and it confirms their mod-3 Gaussian structure at 480 copies. For the other two channels the largest published values, 0.1183711 and 0.1180673 at $n = 30$, were numerical. Our inputs follow that mod-3 Gaussian structure, are refined by gradient ascent on all amplitudes at $n = 80$ and $n = 240$, and are then zero-padded to more copies. A frozen interval-arithmetic verifier certifies 0.06679861, 0.12464123 and 0.12367528 ($n = 480$, 80, 80). The larger values are certified by a separate 128-bit ball-arithmetic evaluator that represents the channels differently, re-certifies the verifier's final records and lies inside the verifier's enclosure on every input on which both were run; its formulas and enclosure lemmas are given in an appendix. A third, non-rigorous evaluation from another formula reproduces the three values to 4×10^{-15} bits. Antidegradability forces $Q = 0$ for $1/12 \le p \le 1/3$, $0.146447 \le p \le 1/2$ and $1/6 \le p \le 1/2$, so the thresholds lie in $[0.066882, 1/12]$, $[0.127705, 0.146447]$ and $[0.126774, 1/6]$. The inputs were found by a language-model agent working against the frozen verifier. The inputs, the evaluators, their validation suites and a replay script are released with this work.

I. INTRODUCTION

The quantum capacity $Q(\mathcal{N})$ of a channel $\mathcal{N}$ is the largest rate at which it transmits quantum information reliably. By the theorem of Lloyd, Shor and Devetak [1–3],

$$Q(\mathcal{N}) = \lim_{n\to\infty} \frac{1}{n} \max_{\rho} I_c\big(\rho, \mathcal{N}^{\otimes n}\big), \qquad (1)$$
$$I_c(\rho,\mathcal{N}) = S\big(\mathcal{N}(\rho)\big) - S\big((\mathcal{N}\otimes \mathrm{id})(\psi_\rho)\big),$$

where ψ_ρ purifies ρ and I_c is the coherent information [4, 5]. Because the coherent information is superadditive [7–9], the limit cannot be dropped, and Q is unknown even for the qubit depolarizing channel. The noise level at which it vanishes is open as well. One direction is easy to certify, though: a single input with $I_c(\rho,\mathcal{N}^{\otimes n}) > 0$, for any n, proves $Q(\mathcal{N}) \ge I_c/n > 0$.

For the depolarizing channel $\rho \mapsto (1-3p)\rho + p(X\rho X + Y\rho Y + Z\rho Z)$, with p the probability of each Pauli error, the maximally mixed single-letter input gives the hashing bound, $Q > 0$ for $p < 0.0630965$ [10]. Degenerate codes do better: Shor and Smolin beat the hashing bound by concatenating a repetition code with a random code [6, 7], Smith and Smolin reached total error $3p = 0.19088$ [8], and Fern and Whaley 0.19128 [11]. Neural-network states and systematic searches over Pauli channels [12–14] did not improve the depolarizing value [15]. Agarwal *et al.* optimised permutation-symmetric rank-two inputs on up to 60 copies and found thresholds up to $p = 0.064657$ (paper, at $n = 45$) and 0.0649112 (released states) [15], computed in floating point. Krohn-Grimberghe certified a 45-copy input of this kind at $p = 0.064956$ with an exact verifier, and certified the best released state of Ref. [15] positive at $p = 0.064912$ [16]. From above, the channel is antidegradable, so $Q = 0$, for $p \ge 1/12$ [22, 23].

Ahmed *et al.* [17] claim a larger lower bound, $p_{\rm th} \ge 0.0676387$ per Pauli ($3p = 0.2029161$ in their total-error convention), and call it the best known. Their argument has two parts. The first is a theorem: on inputs supported on a fixed number K of Dicke excitations, the coherent information of M copies of the depolarizing channel differs from that of a Gaussian thermal attenuator by at most $\Delta_{M,K}$, and $\Delta_{M,K} \to 0$ as $M \to \infty$; the Supplemental Material gives $\Delta_{M,K}$ explicitly. The second is numerical: for one rank-two input with real amplitudes on Fock levels $\equiv 0$ and $\equiv 1 \pmod 3$, supported on levels up to 220, with the output truncated at 320 photons, they report a Gaussian coherent information of about 10^{-12} at this noise level. The bound follows for all sufficiently large M. No M is given at which $\Delta_{M,K}$ is below 10^{-12}, so the argument does not name a number of copies; by our reading of their Eq. (IV.32) the leading term of $\Delta_{M,K}$ decays only like $(\log M/M)^{1/4}$, so the M it would guarantee is far beyond any direct evaluation. The paper reports neither the arithmetic nor an error bound for the evaluation of the 10^{-12} margin. We have no reason to doubt the value. If the numerical step is accepted, their bound is the largest known lower bound on

the depolarizing threshold, and the value certified here is below it.

Smith and Smolin [8] began the study of thresholds for two further families: the independent X–Z channel, with independent bit and phase flips of probability p each (also called the BB84 channel), and the two-Pauli channel, with an X or a Z error of probability p each and no Y errors. Fern and Whaley obtained $Q > 0$ up to $p = 0.11277$ (X–Z) and $p = 0.11398$ (two-Pauli) with concatenated codes [11]. Bhalerao and Leditzky improved both with permutation-invariant codes on $n = 18$ and $n = 24$ copies [18], and Agarwal *et al.* reached 0.1183711 and 0.1180673 at $n = 30$, with the note that the optimiser had not peaked [15]. All of these values are numerical. We know of no certified value for either family and of no published value above those of Ref. [15] (literature searched up to 10 November 2026). The thresholds discussed here concern the quantum capacity. They should not be confused with the bit-error rates up to which the BB84 protocol yields a secret key with one-way post-processing, about 12.4% to 12.9% [19, 20]: those are key-rate thresholds obtained with noisy preprocessing, and they do not bound Q.

This paper gives certified lower bounds on all three thresholds (Table I). The depolarizing certificate at $p = 0.06688201$, total error 0.20064603, is the largest value at which $Q > 0$ has been certified with an explicit input at an explicit n. It moves the certified value past 20% total error, exceeds the previous certified value by 1.93×10^{-3}, and covers more than 70% of the distance from that value to the bound claimed in Ref. [17], which it does not reach. In that sense it is an independent finite-n confirmation of the structure found there, not a new record for the depolarizing threshold. For the independent X–Z and two-Pauli channels the certified values exceed the largest published ones, which are numerical, by 9.3×10^{-3} and 8.7×10^{-3}, and they are the first certified values for these families that we know of. Each value is certified by an explicit rank-two input on n qubits, specified by double-precision amplitudes in the Dicke basis that are taken as exact, together with an exact rational p. The sign of the coherent information is certified in interval arithmetic by one or both of two evaluators that share no code (Sec. IV); the three largest values rest on the second evaluator alone, whose soundness argument is given in Appendix B. The depolarizing and independent X–Z channels form semigroups in p, so a certificate at p proves $Q > 0$ on all of $[0, p]$. The two-Pauli certificate holds at its own p.

Section II reduces the coherent information of symmetric inputs to blocks of polynomial size. Section III describes how the inputs were built, Sec. IV how they were certified, and Sec. V the results. Section VII describes the role of the language-model agents.

II. COHERENT INFORMATION OF SYMMETRIC INPUTS

A qubit Pauli channel acts as $\mathcal{N}(\rho) = \sum_{P\in\{I,X,Y,Z\}} p_P\, P\rho P$. The three families are listed in Table I. We fix the domain of the parameter and the meaning of "threshold" for each.

Depolarizing, $0 \le p \le 1/4$. The channel is $\mathcal{N}_p(A) = (1-4p)A + 2p\,\mathrm{tr}(A)\mathbb{1}$, and $\mathcal{N}_{p'} = \mathcal{N}_q \circ \mathcal{N}_p$ with $1-4p' = (1-4q)(1-4p)$. Every $p' \in [p, 1/4]$ arises with some $q \in [0, 1/4]$, and Q cannot increase under post-processing, so Q is nonincreasing on $[0, 1/4]$. (For $1/4 < p \le 1/3$ the channel is antidegradable, Appendix A.)

Independent X–Z, $0 \le p \le 1/2$. A bit flip and a phase flip are applied independently, each with probability p. Composing the channels at p and q gives the channel at $p' = p + q - 2pq$, and every $p' \in [p, 1/2]$ arises with some $q \in [0, 1/2]$, so Q is nonincreasing on $[0, 1/2]$. The family is defined for all $p \in [0, 1]$, but the channel at $1 - p$ is the channel at p followed by the unitary Y, so $Q(1-p) = Q(p)$; in particular the channel at $p = 1$ is unitary and has $Q = 1$. Statements below are for $p \le 1/2$.

Two-Pauli, $0 \le p \le 1/2$ (the whole family). The composition of two such channels has a Y component, so the family is not closed under composition, and we know of no proof that Q is monotone in p.

For each family we call $p_{\mathrm{th}} = \sup\{p \text{ in the domain} : Q(\mathcal{N}_p) > 0\}$ the threshold. For the depolarizing and X–Z families, monotonicity and the continuity of Q [21] give $Q > 0$ exactly on $[0, p_{\mathrm{th}})$, and a certificate at p proves $Q > 0$ on all of $[0, p]$. For the two-Pauli family a certificate at p proves $Q > 0$ at that p and hence $p_{\mathrm{th}} \ge p$; it says nothing about smaller p. (Both evaluators accept $0 < p \le 1/4$ only, which covers every value used here.)

Inputs. An input of rank $r \le 4$ on n qubits is given by r vectors $|v_q\rangle \in \mathrm{Sym}^n(\mathbb{C}^2)$, written in the Dicke basis $\{|D_a\rangle\}_{a=0}^n$, where a is the number of ones [24]. Let $\rho = \sum_q |v_q\rangle\langle v_q| / \mathcal{Z}$, with $\mathcal{Z} = \sum_q \||v_q\rangle\|^2$, and let $|\Psi\rangle = \mathcal{Z}^{-1/2} \sum_q |v_q\rangle|q\rangle_R$ be its purification.

Schur–Weyl blocks. Under $SU(2) \times S_n$, $(\mathbb{C}^2)^{\otimes n} = \bigoplus_{k=0}^{\lfloor n/2 \rfloor} V_j \otimes \mathbb{C}^{m_k}$ with spin $j = n/2 - k$ and multiplicity $m_k = \binom{n}{k} - \binom{n}{k-1}$ [25]. The output $\mathcal{N}^{\otimes n}(\rho)$ and the joint output-reference state $(\mathcal{N}^{\otimes n} \otimes \mathrm{id}_R)(\Psi)$ commute with permutations of the n outputs, so

$$\mathcal{N}^{\otimes n}(\rho) = \bigoplus_k \sigma_k \otimes \frac{\mathbb{1}_{m_k}}{m_k},$$
$$(\mathcal{N}^{\otimes n} \otimes \mathrm{id}_R)(\Psi) = \bigoplus_k \tau_k \otimes \frac{\mathbb{1}_{m_k}}{m_k}, \tag{2}$$

with $\sigma_k = \mathrm{tr}_R\, \tau_k$ acting on V_j and τ_k on $V_j \otimes \mathbb{C}^r$. The terms $\mathrm{tr}\,\sigma_k \log m_k$ in the two entropies are equal, and

$$I_c = \sum_{k=0}^{\lfloor n/2 \rfloor} \big[H(\sigma_k) - H(\tau_k)\big], \qquad H(A) = -\,\mathrm{tr}\, A \log_2 A. \tag{3}$$

TABLE I. Lower bounds on the threshold $p_{\text{th}} = \sup\{p : Q > 0\}$ of three one-parameter families of Pauli channels (domains in Sec. II). "Certified before" and "claimed or numerical before" are the largest published values of each kind before this work. The verifier row lists board records certified by the frozen verifier (Sec. IV A), and the Arb row lists values certified by the 128-bit evaluator (Sec. IV B); n is the number of copies. For the depolarizing and X–Z families $Q > 0$ on all of $[0, p]$; for the two-Pauli family $Q > 0$ at the listed p. The last row is where antidegradability forces $Q = 0$ (Appendix A).

	depolarizing	independent X–Z	two-Pauli
(p_I, p_X, p_Y, p_Z)	$(1-3p, p, p, p)$	$\big((1-p)^2, p\bar{p}, p^2, p\bar{p}\big)$	$(1-2p, p, 0, p)$
domain of p	$[0, 1/4]$	$[0, 1/2]$	$[0, 1/2]$
hashing	0.0630965	0.1100279	0.1135461
certified before	0.064956 [16]	none	none
claimed or numerical before	0.0676387[a] [17]	0.1183711 [15]	0.1180673 [15]
this work, verifier (n)	0.06679861 (480)	0.12464123 (80)	0.12367528 (80)
this work, Arb (n)	0.06688201 (480)	0.12770564 (320)	0.12677472 (320)
$Q = 0$ for	$1/12 \le p \le 1/3$	$(1 - 1/\sqrt{2})/2 \le p \le 1/2$	$1/6 \le p \le 1/2$

$\bar{p} = 1 - p$. $(1 - 1/\sqrt{2})/2 = 0.1464466$.

[a] Total error 0.2029161. From a convergence theorem for $n \to \infty$ and a numerical value of about 10^{-12} for a Gaussian coherent information; no explicit n, no stated error bound. It exceeds the values certified here.

The blocks have dimensions $n-2k+1$ and $r(n-2k+1)$, so I_c can be evaluated at several hundred copies.

Block entries. The input is a linear combination of operators $(|u\rangle\langle w|)^{\otimes n}$, and $\mathcal{N}^{\otimes n}$ maps each to $Z_{uw}^{\otimes n}$ with $Z_{uw} = \mathcal{N}(|u\rangle\langle w|)$. On the kth sector, $A^{\otimes n}$ acts as $\det(A)^k \operatorname{Sym}^{2j}(A) \otimes \mathbb{1}_{m_k}$. The blocks therefore follow by coefficient extraction from $\det(Z)^k (x^\dagger Z y)^{2j}$ against the input's coefficients. The frozen verifier evaluates this in two ways. Path 0 handles the depolarizing channel, where $\det Z_{uw} = 2p(1-2p)(w^\dagger u)^2$ and $Z_{uw} = (1-4p)|u\rangle\langle w| + 2p(w^\dagger u)\mathbb{1}$. It reduces the blocks to iterated partial traces of ρ, at cost $O(n^4)$. Path 1 handles any Pauli channel. It applies $\det Z$ as a 3×3 correlation kernel and builds $(x^\dagger Z y)^{2j}$ by dynamic programming, at cost $O(n^6)$.

The Y frame. Let $U = e^{-i\pi X/4}$. Conjugation by U exchanges Y and Z, and $I_c(U^{\otimes n}\rho U^{\dagger\otimes n}, \mathcal{N}^{\otimes n}) = I_c(\rho, \mathcal{N}'^{\otimes n})$ with $\mathcal{N}'(\rho) = U^\dagger \mathcal{N}(U\rho U^\dagger) U$. In this frame the independent X–Z channel becomes $\big((1-p)^2, p(1-p), p(1-p), p^2\big)$ and the two-Pauli channel $(1-2p, p, p, 0)$. Both now have $p_X = p_Y$, so they are covariant under rotations about z, like the depolarizing channel in its own frame. We search for X–Z and two-Pauli inputs in this frame, with real amplitudes, and map them back with $U^{\otimes n} = e^{-i\pi J_x/2}$ on $\operatorname{Sym}^n(\mathbb{C}^2)$ before certification.

III. CONSTRUCTION OF THE INPUTS

Ansatz. Ahmed *et al.* found that the many-copy optimum is a rank-two input whose two codewords occupy Fock levels $\equiv 0$ and $\equiv 1 \pmod 3$, with real amplitudes and a Gaussian envelope [17]. We use the same structure in the Dicke basis:

$$v_r(a) \propto \sqrt{q_r}\, g(a)\, [a \equiv r \pmod 3], \quad r = 0, 1, \qquad (4)$$

with $q_0 = q$, $q_1 = 1 - q$, $g(a) = \exp[-(a - a_0)^2/4s^2]$ and $s = s_f\sqrt{a_0}$. Each row is normalised before it is weighted, so the ansatz has three parameters (a_0, s_f, q). For the depolarizing channel the codewords sit at the z pole as written. For the other two channels Eq. (4) is used in the Y frame. Typical optimal parameters are $(a_0, s_f, q) = (36.31, 1.324, 0.5106)$ for the depolarizing channel at $n = 240$, and $(29.16, 1.300, 0.4750)$ (X–Z) and $(29.46, 1.290, 0.4389)$ (two-Pauli) at $n = 80$.

A margin objective. The objective is p_δ, the largest p at which $I_c \ge \delta$, with a margin δ of 2×10^{-9} to 10^{-8} bits, rather than the zero crossing of I_c. Maximising the zero crossing drifts into states whose coherent information stays near 10^{-12} bits over a wide range of p. Such states are valid in principle, but they cannot be certified, because the enclosures are about 10^{-9} bits wide. The ansatz parameters were chosen by Nelder–Mead [26] on p_δ.

Gradient climb. The coherent information has an exact gradient with respect to all amplitudes, obtained from the adjoints of the block maps. L-BFGS [27] maximises I_c at a p slightly above the current p_δ, alternating with updates of p_δ. At $n = 150$ this raised the certified value of the ansatz optimum from 0.06629494 to 0.06645409. At $n = 240$, three rounds of 400 iterations raised the margin threshold ($\delta = 4 \times 10^{-9}$) from 0.06646040 to 0.06659353, 0.06662527 and 0.06663047, after which it levelled off. In the Y frame at $n = 80$, three rounds of 200 iterations ($\delta = 10^{-8}$) raised it from 0.12433724 to 0.12459519 (X–Z) and from 0.12336247 to 0.12363278 (two-Pauli).

Transplanting to more copies. The climbed states are supported on low Dicke levels, $a \lesssim 80$. There $\operatorname{Sym}^n(\mathbb{C}^2)$ looks like a bosonic Fock space: $J_- \approx \sqrt{n}\, b$ by the Holstein–Primakoff approximation [28], which is the regime of the Gaussian limit of Ref. [17]. The same amplitudes, padded with zeros, therefore stay near-optimal at larger n, and they beat a fresh ansatz

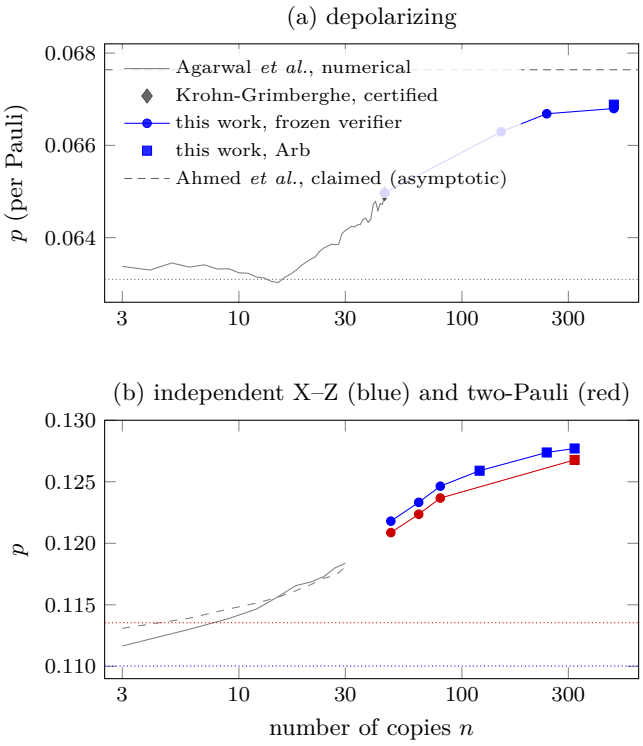


FIG. 1. Certified lower bounds on the thresholds against the number of copies. Circles: certified by the frozen verifier (board records, and at $n = 45$ the re-optimised input of Ref. [16]). Squares: certified by the 128-bit Arb evaluator only, where the frozen verifier cannot decide: beyond its size limit (X–Z and two-Pauli, $n > 80$) or below its resolution (depolarizing, $n = 480$, which is within its size limit). The X–Z point at $n = 120$ is the cross-check of Sec. IV C. Gray lines: numerical thresholds of the best inputs of Ref. [15] (in (b), solid X–Z and dashed two-Pauli). Dotted lines: hashing bounds. Dashed line in (a): the bound 0.0676387 claimed in Ref. [17] for $n \to \infty$, which is above all values certified here. $Q = 0$ from $p = 1/12$, 0.1464 and 1/6 to the end of each domain, above the plotted ranges.

there. Padded to $n = 360$ and 480, the climbed depolarizing state from $n = 240$ reaches margin thresholds 0.06675043 and 0.06680957. The ansatz with the $n = 240$ parameters reaches only 0.06657310 and 0.06662870. A further gradient climb at $n = 360$ gained 2×10^{-6}, so the padded state is close to a local optimum. The $n = 80$ X–Z and two-Pauli records, padded in the Y frame, have float zero crossings 0.12605577, 0.12672929, 0.12738956, 0.12771564 (X–Z) and 0.12513919, 0.12581086, 0.12646389, 0.12678472 (two-Pauli) at $n = 120$, 160, 240 and 320.

The thresholds rise with n at every size we tried, but the gains shrink (Fig. 1). For the depolarizing channel the margin threshold gains 1.2×10^{-4} from $n = 240$ to 360 and 5.9×10^{-5} from 360 to 480. For the X–Z channel the zero crossing gains 6.6×10^{-4} from 160 to 240 and 3.3×10^{-4} from 240 to 320.

IV. CERTIFICATION

A. The frozen verifier

The verifier (`verify.py` with a C engine, `icball.c`) takes an input file with the channel family, a rational p and the amplitude rows, and certifies the sign of Eq. (3).

1. *Exact inputs.* The amplitudes are exactly the double-precision numbers given, p is an exact rational with denominator at most 10^{15}, and the normalisation $\mathcal{Z}$ is exact. Combinatorial scalars are computed exactly or to 40 digits.

2. *Ball arithmetic.* Every block entry is a centre and a radius, with explicit rounding terms for each operation and radii rounded upward.

3. *Eigenvalue enclosures.* A floating-point eigendecomposition is only a hint. Its residual $AQ - Q\Lambda$ and its orthogonality defect $Q^{\mathsf{T}}Q - \mathbb{1}$ are bounded with error-free dot products [29]. A polar-factor argument and Weyl's inequality [30] then bound the distance between the sorted true and computed eigenvalues. Complex Hermitian blocks are embedded as real symmetric matrices of twice the size.

4. *Entropy enclosures.* $-x \ln x$ is bounded on each eigenvalue interval with directed rounding in decimal arithmetic, whose logarithm is correctly rounded.

5. *Alarms.* The verifier stops with an alarm if the total block trace differs from 1, if $\operatorname{tr} \sigma_k$ and $\operatorname{tr} \tau_k$ disagree, if an eigenvalue is certified negative, if paths 0 and 1 disagree (every depolarizing input with $n \le 48$ is evaluated both ways), or if it certifies $I_c > 0$ at a p where its record file lists $Q = 0$. The last alarm is live only for the depolarizing channel ($p \ge 1/12$). For the other two families the frozen record file lists the entanglement-breaking points, $1 - 1/\sqrt{2}$ and 1/4, and the verifier accepts only $p \le 1/4$, so that alarm can fire only for a two-Pauli input at exactly $p = 1/4$; the tighter antidegradability points of Appendix A were not frozen in.

The eigenvalue bound of step 3 is Lemma 1 of Appendix B, where the entropy rule is also stated. The verifier's tests compare both paths with a full density-matrix computation for $n \le 4$ (all three families, ranks 2 and 3), and they reproduce the four certified signs of Ref. [16]: two 45-copy states, each certified positive at one p and negative at a larger one. The verifier was frozen and pinned by hash before the search. Its only revision (version 2) raised the size limit for the depolarizing channel from $n = 240$ to 480. After it, the tests were re-run and the hashes re-pinned. The limits are $n \le 480$ for the depolarizing channel (path 0) and $n \le 80$ for the others (path 1). A run takes 29 s at $n = 240$ and about 10 min

and 1.2 GB at $n = 480$, and 4–7 min for X–Z and two-Pauli inputs at $n = 80$. At $n = 120$ path 1 takes 43 min, and its enclosure, 5.3×10^{-8} bits wide, is too wide to be useful. The pinned hashes (first 16 hexadecimal digits of the SHA-256) are `4f007e83056ea3a1` (`verify.py`), `00b09294b432e40a` (`icball.c`) and `3901d412c50f84f7` (record file).

B. An independent 128-bit evaluator

The second evaluator shares no code with the verifier and does not use its kernel. It writes each Pauli channel as e^G with $G = \sum_{P \neq I} \gamma_P (P \cdot P - \mathrm{id})$, where the rates γ_P follow from the logarithms of the channel's Pauli eigenvalues, and lets G act on the Schur–Weyl blocks of Sec. II through spin-$\frac{1}{2}$ Clebsch–Gordan coefficients. This is a Dicke master equation, as in Ref. [31]. The part with positive rates is applied by Poisson uniformisation with a rigorous trace-norm bound on the truncated tail. The bound was at most 3×10^{-44} in every run, and at most 3×10^{-37} when the Taylor tail below is included. The two-Pauli channel has one negative rate, and that commuting factor is applied by a Taylor series with a rigorous tail. The X–Z and two-Pauli channels are evaluated in the Y frame, using an exact Krawtchouk closed form for the matrix elements of $e^{-i\pi J_x/2}$. All arithmetic is Arb ball arithmetic [32] at 128 bits, starting from the exact inputs. Spectra start from floating-point hints refined in Arb, which are not trusted. They are certified by Gershgorin discs of $X^{\mathsf{T}} H X$, with Ostrowski's theorem [30] covering the non-orthogonality of the hint basis X, and $-x \log_2 x$ is bounded on each interval. The cost is $O(n^4)$ for every Pauli channel. Appendix B gives the generator, its action on the blocks, the tail bound and the spectral enclosure with their proofs, and the versions and hashes of the code.

The evaluator was validated before it was used on new inputs:

- it encloses brute-force values in 252 cases with $n \leq 8$, covering all families and ranks 1–3, and a 50-digit reference in 35 further cases;
- its spectral routine passed a unit test with exact degeneracies and eigenvalue clusters of width 10^{-30} to 10^{-76};
- on the 45-copy states of Ref. [16] its values lie inside Krohn-Grimberghe's certified intervals, and it reproduces all four of their signs;
- it returns a negative sign for two controls: a p just above a state's zero crossing, and a state with one amplitude perturbed by 10^{-4};
- with its series cut far earlier than in production, so that the truncation bound of Lemma 2 dominates the enclosure (bounds from 10^{-30} up to 10^{-6}), it still encloses the 50-digit reference in 30 cases and, at $n = 80$, the production enclosures of the final X–Z and two-Pauli records in 4 cases. This last test was added in the revision of November 2026; the evaluator was not changed.

Unlike the verifier, the evaluator is not a frozen board component. It was written after the search at the original size limits had finished, and its values are reported separately from the board records (Table III). The three largest values of this paper are certified by this evaluator alone. For them the paper is a computer-assisted proof whose correctness rests on the argument of Appendix B, on this code (three files, about 1100 lines, identified by hash in Table IV), and on Arb.

C. Agreement

Table II lists the final board records with both evaluations. The Arb value lies inside the verifier's enclosure in every case. The other 17 board records (Appendix D) were evaluated with Arb in the revision of November 2026: all are positive, and each Arb enclosure lies inside the verifier's. Both evaluators were also run on the three points with margin of Sec. V, on the 45-copy states, on the depolarizing input of Table III and on one input above the verifier's size limit. On the depolarizing input of Table III ($n = 480$, $p = 0.06688201$) the verifier's enclosure $[-4.0218, 2.8387] \times 10^{-9}$ contains zero, so it decides nothing there, and it contains the Arb value 4.852×10^{-10}. Above the verifier's two-Pauli and X–Z limit, the $n = 80$ X–Z record padded to $n = 120$, at $p = 0.1259$, gives an inconclusive verifier enclosure $[-6.2267 \times 10^{-9}, 4.6651 \times 10^{-8}]$ (43 min), a floating-point value 2.7602×10^{-8}, and the Arb value $2.7601677157 \times 10^{-8}$. The verifier's general kernel and the master-equation representation therefore agree at $n = 120$ as well.

Two further checks were run during the search. First, an independent floating-point evaluator, using the same master-equation representation as the 128-bit evaluator, was validated against brute force for $n \leq 10$ (to 1.7×10^{-14}), against Krohn-Grimberghe's exact certificates (to 1.3×10^{-15}), and against a 45-digit re-run at $n = 48$. It placed every board record up to #28 inside the verifier's enclosure. Second, Krohn-Grimberghe's verifier [16], run unmodified, accepted our re-optimised 45-copy input at $p = 6497/100000$ with $I_c = 5.3454324111 \times 10^{-9}$ bits. This value lies inside our verifier's enclosure $[5.2932, 5.3929] \times 10^{-9}$. As a calibration, the certificate our emitter produced for their own state was accepted with a value inside their published interval, and their verifier rejected all five tampered certificates. Their code is specific to 45 copies, rank two and the depolarizing channel, so it confirms our kernel and entropy bounds but not the larger inputs directly.

TABLE II. Final board records: the frozen verifier's enclosure of I_c and the Arb value. The Arb enclosures are about 10^{-28} bits wide, and the digits shown are exact (lower end, truncated). Board posts #20, #30, #28 and #27.

channel	n	p	verifier I_c (bits)	Arb I_c (bits)
depol.	240	0.0666847	[9.3916e−12, 2.1094e−9]	1.3129797632e−9
depol.	480	0.06679861	[2.4950e−11, 6.8873e−9]	4.5321740991e−9
X–Z	80	0.12464123	[3.5713e−11, 2.6056e−9]	1.6050687056e−9
2-Pauli	80	0.12367528	[1.3223e−13, 7.5729e−9]	4.6880049254e−9

TABLE III. Values certified by the 128-bit evaluator where the frozen verifier cannot decide: beyond its size limit ($n \leq 80$) for the X–Z and two-Pauli inputs, and below its resolution for the depolarizing input, which is within its size limit ($n \leq 480$) but on which it returns UNDECIDED, $I_c \in [-4.0218, 2.8387] \times 10^{-9}$ bits. The inputs are the board records zero-padded to n copies. I_c is the lower end of the enclosure, truncated. "Time" is the wall-clock time of the original run with the given number of processes.

channel	n	p	I_c (bits)	time
depol.	480	0.06688201	4.8519099129e−10	2.9 h, 3
X–Z	240	0.12737956	1.7306652728e−9	53 min, 4
X–Z	320	0.12770564	1.7220458670e−9	2.2 h, 4
2-Pauli	320	0.12677472	1.1898001345e−9	2.9 h, 3

D. A second evaluation of the largest values

Because the values of Table III rest on one certifying program, we evaluated the same inputs once more with a program that shares neither code nor channel representation with it (Appendix C). It builds the blocks from the generating function $\det(Z)^k \operatorname{Sym}^{2j}(Z)$ by a recurrence in the spin with positive terms only, uses exact integers for the change of frame, and takes eigenvalues from LAPACK in double precision. It returns a number, not an enclosure, and *it is not rigorous*: it corroborates the certified signs and does not replace the certificates. Against a full density-matrix computation for $n \leq 7$ (126 inputs, all families, ranks 1–3) its largest deviation is 8×10^{-15} bits. Run in 30-digit arithmetic it reproduces the Arb values at the four certified points of the 45-copy states of Ref. [16], of our 45-copy input, of the two final $n = 80$ board records and of the X–Z input at $n = 120$ of Sec. IV C to within 3×10^{-30} bits, that is, to 21 or more significant digits. The headline inputs are too large for the 30-digit mode, whose cost grows like n^4 with a large constant (17 minutes at $n = 120$). On the inputs of Table III it gives, in double precision, $4.8518710 \times 10^{-10}$ (depolarizing, $n = 480$), 1.7220446×10^{-9} (X–Z, $n = 320$), 1.7306661×10^{-9} (X–Z, $n = 240$) and 1.1898039×10^{-9} (two-Pauli, $n = 320$) bits. These differ from the certified values by 3.9, 1.2, 0.8 and 3.7×10^{-15} bits, five orders of magnitude below the values themselves. At $p + 1.1 \times 10^{-5}$ the same three headline inputs give -4.8×10^{-11}, -1.7×10^{-10} and -1.2×10^{-10} bits, consistent with the zero crossings quoted in Sec. III. The evaluation takes 35 s at $n = 480$. An independent referee of this manuscript obtained the same agreement with a separately written evaluator of the same kind.

E. Choosing p

The large-n inputs are very flat in p. Near the threshold, $|dI_c/dp|$ is about 0.02 bits per unit p at $n = 45$–48, but only 3.5 to 5.7×10^{-5} for the depolarizing channel at $n = 150$ and 240, and 4.9×10^{-5} at $n = 480$. An enclosure of width 10^{-9} bits therefore costs 10^{-6} to 3×10^{-5} in p, and fixed float margins left between 10^{-5} and 7×10^{-5} unused. We therefore chose each board value by bisecting p against the verifier itself, and each Arb value at the float zero crossing minus 10^{-5}, rounded down to a multiple of 10^{-8}. At $n = 480$ the depolarizing zero crossing is 0.06689201. The verifier's enclosure is 6.9×10^{-9} bits wide, mostly from the double-precision floor on many tiny eigenvalues, and it certifies 9.3×10^{-5} below the crossing: in the bisection it was already undecided at $p = 0.06679926$. Arb certifies 10^{-5} below the crossing. This, and not the size limit, is why the depolarizing value of Table III has no verifier certificate.

V. RESULTS

The frozen verifier certifies $Q > 0$ for the depolarizing channel at $p = 0.06679861$, $3p = 0.20039583$, with a rank-two input on 480 qubits; for the independent X–Z channel at $p = 0.12464123$; and for the two-Pauli channel at $p = 0.12367528$, both on 80 qubits (Table II). Where the verifier cannot decide, the 128-bit evaluator certifies more: the depolarizing channel at $p = 0.06688201$, $3p = 0.20064603$, on 480 qubits (within the verifier's size limit, below its resolution), and the X–Z channel at $p = 0.12770564$ and the two-Pauli channel at $p = 0.12677472$, both on 320 qubits (beyond its size limit) (Table III). The depolarizing values exceed the previous certified value, 0.064956 [16], by 1.84×10^{-3} (verifier) and 1.93×10^{-3} (Arb), and they are 8.4×10^{-4} and 7.6×10^{-4} below the bound claimed in Ref. [17]. The X–Z and two-Pauli values exceed the numerical thresholds of Ref. [15] by 6.27×10^{-3} and 5.61×10^{-3} (verifier), and

by 9.33×10^{-3} and 8.71×10^{-3} (Arb). In total, the board holds 21 records, listed in Appendix D.

Calibration. The verifier certifies the released $n = 30$ states of Ref. [15] positive at $p = 0.11837$ (X–Z) and 0.118066 (two-Pauli), and negative at 0.118372 and 0.118068. These are the first certified values for those states, and they agree with the published numerical thresholds.

Points with margin. The records sit at the edge of what each evaluator accepts: the verifier's lower ends lie between 10^{-13} and 4×10^{-11} bits. Statements with more margin are available. These are the largest p at which the verifier's lower bound is at least 10^{-9} bits, each also certified with Arb: the depolarizing channel at $p = 0.06666451$ ($n = 240$, Arb $I_c = 2.3132493724\times10^{-9}$), the X–Z channel at 0.12463577 ($n = 80$, verifier $[1.0227, 3.6058]\times10^{-9}$, Arb $2.6004754111\times10^{-9}$) and the two-Pauli channel at 0.12366746 ($n = 80$, verifier $[1.0017, 8.5360]\times10^{-9}$, Arb $5.6654186637\times10^{-9}$).

Open windows. Antidegradability forces $Q = 0$ for $1/12 \le p \le 1/3$ (depolarizing), $(1-1/\sqrt{2})/2 = 0.1464466 \le p \le 1/2$ (X–Z) and $1/6 \le p \le 1/2$ (two-Pauli) (Appendix A). With the certified values, the thresholds p_{th} of Sec. II lie in $[0.06688201, 1/12]$, $[0.12770564, 0.1464467]$ and $[0.12677472, 1/6]$, windows of widths 0.0165, 0.019 and 0.040. For the depolarizing channel the certified window is 10% narrower than with the previous certified value; if the bound of Ref. [17] is accepted, its lower end moves to 0.0676387.

VI. DISCUSSION

The certified depolarizing value, $3p = 0.20064603$, is 2.3×10^{-3} below the value 0.2029161 claimed in Ref. [17], which is not certified at any explicit n. The two results are complementary. Theirs explains the structure of the good inputs and, given its numerical step, gives the larger bound; ours shows with certified arithmetic that inputs of that structure have positive coherent information at 480 copies, at 98.9% of their noise level. The gains with n are shrinking, so closing the gap with the present construction would need inputs on well over 480 qubits. A certified evaluation of the Gaussian coherent information of Ref. [17], together with an explicit M from their $\Delta_{M,K}$, would be another route. The Arb evaluator and path 0 of the verifier scale as $O(n^4)$, path 1 as $O(n^6)$. At $n = 480$ the Arb evaluation took 2.9 hours on three processes, and the verifier's enclosure is already widened by double-precision rounding. For the X–Z and two-Pauli channels the thresholds were still rising by a few 10^{-4} per 100 copies at $n = 320$. We do not know their many-copy limits.

The certificates prove positivity, not useful rates: the certified coherent information is of order 10^{-9} bits per block of several hundred channel uses. All our inputs have rank two. The verifier accepts rank up to four, and higher rank and other residue structures are unexplored.

The construction combines three ingredients: the mod-3 Gaussian structure predicted by the many-copy analysis [17], a gradient climb on all amplitudes with a certification margin built into the objective, and zero-padding to more copies. The last is the cheapest. Padding a state climbed at $n = 240$ or $n = 80$ raises its margin threshold by between 1.2×10^{-4} and 3.1×10^{-3} with no further search, which is consistent with the Holstein–Primakoff picture of the low Dicke levels.

For the X–Z (BB84) and two-Pauli channels, the values of Fern and Whaley (0.11277 and 0.11398) [11], of Bhalerao and Leditzky [18] and of Agarwal *et al.* (0.1183711 and 0.1180673) [15] are all numerical, and all lie below the certified values of Table I. Bhalerao and Leditzky use the two-Pauli convention $(1-p)\rho + \frac{p}{2}(X\rho X + Z\rho Z)$, with total error p, and report their thresholds in figures only; the noise levels they quote in the text (two-Pauli total error 0.2271, i.e. 0.11355 per Pauli, and BB84 0.112105) are those of example codes, not thresholds.

VII. HOW THE RESULTS WERE FOUND

The verifier, its tests and the record snapshot (25 October 2026) were built by a coordinating language-model session (Claude Opus 5.5, Anthropic). The same session then carried out the search as the only agent on the problem's message board, from 26 to 28 October 2026. A planned three-agent swarm was not launched. Only verdicts of the frozen verifier counted as board records: the verifier was read-only and pinned by hash, and the board accepts a verdict as a record only if the pinned hashes match. Because the searching session had also written the verifier, every record was re-checked with code written separately by subagents that the session launched. One subagent re-ran Krohn-Grimberghe's verifier on the 45-copy input, one wrote the independent floating-point evaluator, and one built and validated the 128-bit evaluator. The first three board records were posted between 21:09 and 21:11 on 26 October, when the session joined the board. The depolarizing record at $n = 240$ followed at 05:39 on 27 October, and the X–Z and two-Pauli records at $n = 80$ at 12:52 and 12:55. After the verifier's size limit was raised, the $n = 480$ record followed at 02:54 on 28 October. The last Arb evaluation beyond the limits finished at 05:50 that morning. The human operator decided that the 128-bit evaluator would stay a paper-level certifier, separate from the board. Apart from the size-limit revision, the verifier and the record data were not changed, and the 128-bit evaluator was not changed after its validation. This manuscript was drafted by a language model (Claude Opus 5.5) from the board, the audit records and the evaluation logs, and checked against them. Two reviews followed, on 31 October and 9–10 November 2026; the second re-ran the frozen verifier on all 21 board records and on the depolarizing input of Table III, and confirmed the values of Table III with its own floating-point evaluator. The present revision

answers both. It was prepared by a further language-model session (Claude Fable 5.2, Anthropic), which also wrote the second evaluator of Sec. IV D, assembled the release and re-ran the certificates as recorded there. No certificate, input or certifying program was changed in the revision.

ACKNOWLEDGMENTS

We thank A. Krohn-Grimberghe for releasing the states, certificates and verifier of Ref. [16] under an open license (CC BY 4.0), and A. Agarwal, A. R. Kalra, S. Lee, D. Leung, L. Schaeffer, P. Sinha and G. Smith for publishing their states and threshold tables [15]. *Use of AI.* The inputs were found and certified by Claude Opus 5.5 (Anthropic), which also wrote the verification software and the first version of this manuscript; the revision and the second evaluator were written by Claude Fable 5.2 (Anthropic). Section VII gives the details.

DATA AVAILABILITY

The release that accompanies this revision is version 2 of the artifact (tag `capacity-thresholds-v2`), at https://github.com/maxxfan/aixmathxphysics/tree/capacity-thresholds-v2/papers/capacity-thresholds/artifacts. Version 1 of the artifact, at https://github.com/maxxfan/aixmathxphysics/tree/capacity-thresholds-v1/papers/capacity-thresholds/artifacts, contains the 21 board records, the inputs of Table III and the points with margin, the frozen verifier with its tests and pinned hashes, the three files of the 128-bit evaluator with the logs of its runs, and the audit record. All 66 of its data and code files are unchanged in version 2. Version 2 adds what a reader needs to check the evaluator and not only to run it: its validation suite with the test inputs and logs (brute force, 50-digit references, the spectral unit test, the Krohn-Grimberghe states, the negative controls), the floating-point master-equation evaluator and the tools used with Krohn-Grimberghe's verifier, the second evaluator of Sec. IV D, the search programs with their logs as they were run, expected outputs, pinned versions (Table IV), a script that regenerates the release from the project files, and a replay script. The replay script has a quick mode (about 10 minutes: checksums, the verifier's tests and one board record, the validation suite, the two $n = 80$ records with the 128-bit evaluator, the second evaluator on the inputs of Table III), an extended mode, and a mode that re-runs the three largest certificates with one command each (2 to 3 hours each). Each step is compared with its expected output. The release includes the outputs of the replay of 10 November 2026, run from fresh copies on the machine of Table IV: the three largest certificates were reproduced with identical enclosures (3.4, 1.3 and 1.7 hours), as were all other enclosures that were re-computed. Running `python3 verifier/verify.py` on an input re-certifies a board record, and `python3 arb/arb_ic.py` on an input re-runs the 128-bit evaluation.

Appendix A: Where $Q = 0$

A channel is antidegradable, and so has $Q = 0$, if its Choi state has a symmetric extension on the output side [33]. The Choi state of a Pauli channel is Bell-diagonal with weights (p_I, p_X, p_Y, p_Z) and maximally mixed marginals. For two-qubit states, a symmetric extension exists if and only if $\operatorname{tr}\rho_B^2 \geq \operatorname{tr}\rho_{AB}^2 - 4\sqrt{\det\rho_{AB}}$ [33, 34], which here reads $\frac{1}{2} \geq \sum_P p_P^2 - 4\sqrt{p_I p_X p_Y p_Z}$. On the domains of Sec. II this gives the following.

- Depolarizing: the condition is $\frac{1}{2} \geq 1 - 6p + 12p^2 - 4p\sqrt{p(1-3p)}$. It holds exactly for $1/12 \leq p \leq 1/3$, so $Q = 0$ on $[1/12, 1/4]$ and beyond. This is the no-cloning bound [22, 23].
- Independent X–Z: with $t = p(1-p)$ the condition is $\frac{1}{2} \geq 1 - 4t$, i.e. $p(1-p) \geq 1/8$. It holds exactly for $(1 - 1/\sqrt{2})/2 \leq p \leq (1 + 1/\sqrt{2})/2$, so $Q = 0$ on $[(1-1/\sqrt{2})/2,\ 1/2]$. It fails for $p > (1+1/\sqrt{2})/2 = 0.8536$, where $Q(p) = Q(1-p)$ and the certificates of this paper give $Q > 0$ for $p \geq 1 - 0.12770564$.
- Two-Pauli: the determinant vanishes and the condition is $12p^2 - 8p + 1 \leq 0$. It holds exactly for $1/6 \leq p \leq 1/2$, so $Q = 0$ on $[1/6, 1/2]$.

On the part of each domain where p_I is the largest weight, the condition is equivalent to $\sqrt{p_I} \leq \sqrt{p_X} + \sqrt{p_Y} + \sqrt{p_Z}$. A semidefinite program for the extension gives the same three lower ends (0.083333, 0.146446 and, with its tolerance of 10^{-7}, 0.166545). For the X–Z and two-Pauli families these points are tighter than the entanglement-breaking points, $1 - 1/\sqrt{2} = 0.293$ and $1/4$, which are the ones in the verifier's frozen record file (Sec. IV A).

Appendix B: Soundness of the two certifying evaluators

Both evaluators receive the same exact data: the amplitudes, which are double-precision numbers and hence dyadic rationals, the rational p, and the exact normalisation $\mathcal{Z} = \sum_q \|v_q\|^2$. Each returns an interval that is claimed to contain I_c of Eq. (3). This appendix states what each computes and the inequalities that make the interval valid. The blocks σ_k and τ_k are those of Sec. II: they are normalised so that $\operatorname{tr}\sigma_k = \operatorname{tr}\tau_k$ is the probability of sector k ("weighted" blocks, m_k times the operator that acts on one copy of V_j), and I_c is given by Eq. (3). All blocks are positive semidefinite, being blocks of the image of a positive operator under a completely positive map.

1. The frozen verifier

Ball arithmetic. Every quantity computed by the C engine is a pair (c, r) of doubles standing for the set of reals within r of c. Each operation computes the centre in round-to-nearest and adds to the radius the bound $u|c|$, $u = 2^{-53}$, on its rounding error, plus 2^{-1074} where underflow is possible; radii are propagated with every operation rounded upward. The engine is compiled without floating-point contraction. The scalars of the block formula (binomial coefficients, multiplicities, powers of the channel parameters) are computed in exact rational arithmetic or to 40 decimal digits and passed to the engine as balls. The block entries are the coefficient extractions of Sec. II, evaluated in this arithmetic, so the true block lies in the ball matrix.

Lemma 1. *Let A be a real symmetric $m \times m$ matrix, Q a real $m \times m$ matrix and $\Lambda = \mathrm{diag}(\tilde\lambda_1, \dots, \tilde\lambda_m)$. Put $W = AQ - Q\Lambda$ and $E = Q^{\mathsf T}Q - \mathbb{1}$, and let $\eta \geq \|E\|_F$ with $\eta < 1$. Then, with both spectra sorted,*

$$|\lambda_i(A) - \tilde\lambda_i| \leq \frac{\|W\|_F + 2\eta \max_j |\tilde\lambda_j|}{1 - \eta} \qquad \text{for all } i. \tag{B1}$$

Proof. Write $Q = UH$ with U orthogonal and $H = (Q^{\mathsf T}Q)^{1/2}$. The eigenvalues of $Q^{\mathsf T}Q$ lie in $[1-\eta, 1+\eta]$, so those of H lie in $[1-\eta, 1+\eta]$ as well, $\|H - \mathbb{1}\|_2 \leq \eta$ and $\|H^{-1}\|_2 \leq 1/(1-\eta)$. From $AQ = Q\Lambda + W$ we get $U^{\mathsf T}AU = H\Lambda H^{-1} + U^{\mathsf T}WH^{-1}$, and $H\Lambda H^{-1} - \Lambda = [(H - \mathbb{1})\Lambda - \Lambda(H - \mathbb{1})]H^{-1}$. Hence $\|U^{\mathsf T}AU - \Lambda\|_2 \leq (2\eta\|\Lambda\|_2 + \|W\|_2)/(1-\eta)$. The symmetric matrix $U^{\mathsf T}AU$ has the spectrum of A, and Weyl's inequality [30] gives Eq. (B1). □

The verifier applies the lemma to the centre matrix A of each block, with Q and $\tilde\lambda$ from a floating-point eigensolver that is not trusted. It requires $\eta < 1/2$. The entries of W and E are evaluated with the twice-working-precision dot product of Ref. [29], and the a priori error bound of that algorithm, with the underflow terms, is added to $\|W\|_F$ and to η. The true block differs from the centre matrix by at most the Frobenius norm of the radius matrix in spectral norm, and this norm is added to the right side of Eq. (B1) (Weyl again). The result is one radius δ per block with $|\lambda_i - \tilde\lambda_i| \leq \delta$ for all i. A complex Hermitian block B is replaced by the real symmetric matrix with blocks $\mathrm{Re}B, -\mathrm{Im}B, \mathrm{Im}B, \mathrm{Re}B$, whose spectrum is that of B with every eigenvalue doubled; its entropy sum is halved.

Entropy rule. Each eigenvalue lies in $[a, b] = [\max(0, \tilde\lambda_i - \delta), \max(0, \tilde\lambda_i + \delta)]$, with the ends rounded outward; the lower end may be clipped at zero because the block is positive semidefinite, and the verifier stops with an alarm if $\tilde\lambda_i + \delta < 0$. The function $h(x) = -x \ln x$ is concave on $[0, 1]$ with maximum $1/e$ at $x = 1/e$. So $h \geq \min\{h(a), h(b)\}$ on $[a, b]$, and $h \leq h(b)$ if $b < 1/e$, $h \leq h(a)$ if $a > 1/e$, and $h \leq 1/e$ otherwise. These bounds are evaluated in 40-digit decimal arithmetic with directed rounding, using a correctly rounded logarithm widened by one unit in the last place on each side. Summing lower bounds for σ_k and upper bounds for τ_k, and conversely, gives the enclosure of I_c, which is finally divided by $\mathcal{Z} \ln 2$ with outward rounding (the engine works with the unnormalised input).

Consistency. The verifier stops if the trace balls of σ_k and τ_k are disjoint for some k, or if the ball for $\sum_k \mathrm{tr}\,\sigma_k$ does not contain the exact input trace.

2. The 128-bit evaluator

Generator. Let $\lambda_x = p_I + p_X - p_Y - p_Z$, and λ_y, λ_z likewise, be the eigenvalues of the channel on the Pauli matrices; they are positive for $p < 1/4$ in all three families, and the evaluator rejects an input with a nonpositive eigenvalue. For a Pauli matrix P, the map $\rho \mapsto P\rho P - \rho$ has eigenvalue 0 on the Pauli matrices that commute with P and -2 on the others. Hence

$$\mathcal{N} = \exp\Big[\sum_{P \in \{X,Y,Z\}} \gamma_P \,(P \cdot P - \mathrm{id})\Big], \tag{B2}$$

$$\gamma_X = \tfrac{1}{2}(L_y + L_z - L_x) \text{ and cyclic}, \quad L_a = -\tfrac{1}{2}\ln\lambda_a,$$

and $\mathcal{N}^{\otimes n} = e^G$ with $G = \sum_P \gamma_P(\mathcal{S}_P - n)$, $\mathcal{S}_P(Y) = \sum_{i=1}^n P_i Y P_i$. The three $\mathcal{S}_P$ commute. The evaluator works in a frame with $p_X = p_Y$ (Sec. II), where $\gamma_X = \gamma_Y = L_z/2 > 0$ and $\gamma_Z = L_x - L_z/2$. This γ_Z equals γ_X for the depolarizing channel, is zero for the X–Z channel and is negative for the two-Pauli channel. The logarithms are Arb balls computed from the exact rational λ_a.

Action on the blocks. Let $Y = \bigoplus_j Y_j \otimes \mathbb{1}_{m_j}$ commute with permutations, with weighted blocks $\tilde Y_j = m_j Y_j$, and let $C_{jj''} : V_j \to V_{j''} \otimes \mathbb{C}^2$ be the Clebsch–Gordan isometry for $j = j'' \pm \frac{1}{2}$. Splitting off one qubit and averaging over permutations gives, for single-qubit operators A, B,

$$\Big(\sum_{i=1}^n A_i Y B_i\Big)^{\sim}_{j'} = \sum_{j''=j'\pm\frac12} \sum_{j=j''\pm\frac12} \frac{n\, m_{j''}^{(n-1)}}{m_j^{(n)}} \times C^{\dagger}_{j'j''}(\mathbb{1} \otimes A)\, C_{jj''}\, \tilde Y_j\, C^{\dagger}_{jj''}(\mathbb{1} \otimes B)\, C_{j'j''}, \tag{B3}$$

where $m_j^{(N)}$ is the multiplicity of spin j in N qubits (zero if the spin does not occur). The Clebsch–Gordan coefficients are square roots of rationals, $\langle j'', m - \frac{s}{2}; \frac12, \frac{s}{2} | j'' \pm \frac12, m\rangle = \pm\sqrt{(j'' \pm sm + \frac12)/(2j''+1)}$ with the Condon–Shortley signs ($s = \pm 1$), and are evaluated as Arb balls. With $\mathcal{S}_X + \mathcal{S}_Y = 2\sum_i (E_i^{01}\cdot E_i^{10} + E_i^{10}\cdot E_i^{01})$, $E^{01} = |0\rangle\langle 1|$, both $\mathcal{S}_X + \mathcal{S}_Y$ and $\mathcal{S}_Z$ map the Δth diagonal of every block to Δth diagonals, so the diagonals are propagated independently. The matrix elements in Eq. (B3) are real. The real and imaginary parts of the block entries are therefore propagated as separate real vectors, which is exact linear algebra and involves no norm estimate.

Lemma 2. *Let* $\gamma_Z^+ = \max(\gamma_Z, 0)$, $\mu = n(2\gamma_X + \gamma_Z^+)$ *and* $\mathcal{M} = [\gamma_X(\mathcal{S}_X + \mathcal{S}_Y) + \gamma_Z^+ \mathcal{S}_Z]/\mu$. *Then* $e^{G_+} := \exp[\gamma_X(\mathcal{S}_X + \mathcal{S}_Y - 2n) + \gamma_Z^+(\mathcal{S}_Z - n)] = \sum_{k\geq 0} w_k \mathcal{M}^k$ *with* $w_k = e^{-\mu}\mu^k/k!$, *and for every operator* Y *and every* $K > \mu - 2$,

$$\left\| e^{G_+} Y - \sum_{k\leq K} w_k \mathcal{M}^k Y \right\|_1 \leq \epsilon_P \|Y\|_1, \qquad \epsilon_P = \frac{w_{K+1}}{1 - \mu/(K+2)}. \tag{B4}$$

If $\gamma_Z < 0$, *put* $x = 2n|\gamma_Z|$. *Then* $\|e^{\gamma_Z(\mathcal{S}_Z - n)}Y - \sum_{k\leq K'}[\gamma_Z(\mathcal{S}_Z - n)]^k Y/k!\|_1 \leq \epsilon_T \|Y\|_1$ *with* $\epsilon_T = \sum_{k>K'} x^k/k! \leq x^{K'+1}/[(K'+1)!\,(1 - x/(K'+2))]$ *for* $K' > x - 2$, *and applying the truncated Poisson sum to the truncated Taylor sum differs from* $e^G Y$ *by at most* $(\epsilon_T + e^x \epsilon_P)\|Y\|_1$ *in trace norm.*

Proof. $\mathcal{M}$ is a convex combination of the maps $Y \mapsto P_i Y P_i$, so it is a trace-norm contraction, and $G_+ = \mu(\mathcal{M} - 1)$. The tail of the Poisson series is at most $\sum_{k>K} w_k$, which the geometric series bounds by ϵ_P once $\mu/(K+2) < 1$. Next, $\|\mathcal{S}_Z - n\|_{1\to 1} \leq 2n$ gives the Taylor tail, the bound $\|T\|_{1\to 1} \leq e^x$ for the truncated Taylor sum T, and $\|e^{G_+}\|_{1\to 1} \leq 1$. The two factors commute, and $e^G Y - \Sigma_K T Y = e^{G_+}(e^{\gamma_Z(\mathcal{S}_Z - n)} - T)Y + (e^{G_+} - \Sigma_K)TY$ for the truncated Poisson sum Σ_K. □

The evaluator chooses K and K' so that ϵ_P and ϵ_T are each below a target (10^{-45} by default), and uses $\epsilon := \epsilon_T + e^x \epsilon_P$, computed in Arb and rounded up (at most 3×10^{-37} in every run). The lemma is applied to the r^2 operators $X_{qq'} = |v_q\rangle\langle v_{q'}|/\mathcal{Z}$, for which $\sum_{q,q'} \|X_{qq'}\|_1 = (\sum_q \|v_q\|)^2/\mathcal{Z} \leq r$. The exact joint state $\sum_{qq'} e^G(X_{qq'}) \otimes |q\rangle\langle q'|$ therefore differs from the one built from the truncated series by at most $r\epsilon$ in trace norm, and so does its partial trace. The trace norm of a permutation-invariant operator is the sum of the trace norms of its weighted blocks, and the spectral norm is at most the trace norm. By Weyl's inequality every eigenvalue of every exact block is within $r\epsilon$ of the corresponding eigenvalue of the truncated-series block, which lies in the computed ball matrix. The bound is applied to the complex operators as a whole, after the real and imaginary parts have been recombined.

Change of frame. For the X–Z and two-Pauli channels the input is first mapped by $W^{\otimes n}$, $W = (1 - iX)/\sqrt{2}$, using $\langle D_b|W^{\otimes n}|D_a\rangle = 2^{-n/2}\sqrt{\binom{n}{a}/\binom{n}{b}}\,(-i)^{a+b}\sum_c (-1)^c \binom{a}{c}\binom{n-a}{b-c}$, with the integer sum exact and the remaining factors Arb balls. Conjugation by W exchanges p_Y and p_Z.

Lemma 3. *Let* H *be Hermitian* $d \times d$, X *any* $d \times d$ *matrix with* $\|X^\dagger X - \mathbb{1}\|_2 \leq \eta < 1$, *and* $B = X^\dagger H X$ *with Gershgorin intervals* $D_i = [b_{ii} - \rho_i, b_{ii} + \rho_i]$, $\rho_i = \sum_{j\neq i} |b_{ij}|$. *If the union of* κ *of these intervals is an interval* $[\ell, u]$ *disjoint from the other* $d - \kappa$, *then exactly* κ *eigenvalues of* H, *counted in the sorted order with the same indices as the eigenvalues of* B *in* $[\ell, u]$, *lie in* $[\ell/(1 \pm \eta),\ u/(1 \mp \eta)]$, *where the signs are chosen to make the interval largest (upper signs if* $\ell \geq 0$ *and* $u \geq 0$*).*

Proof. Gershgorin's theorem with the usual continuity argument places exactly κ eigenvalues of B in $[\ell, u]$. Ostrowski's theorem [30] gives $\lambda_i(B) = \theta_i \lambda_i(H)$ for the sorted eigenvalues, with θ_i between the smallest and the largest eigenvalue of $X^\dagger X$, hence in $[1 - \eta, 1 + \eta]$. □

The evaluator takes X from a floating-point eigendecomposition refined heuristically in Arb; nothing about X is assumed. It computes B and $X^\dagger X - \mathbb{1}$ in ball arithmetic from the ball matrix H and the exact matrix X, bounds η by the largest absolute row sum of $X^\dagger X - \mathbb{1}$, and requires $\eta < 1/4$. Since the Gershgorin intervals are formed from upper bounds, the conclusion holds for every Hermitian matrix in the ball H. Each resulting interval is widened by $r\epsilon$, clipped at zero from below (a wholly negative interval stops the program), and contributes $\kappa \min\{h(a), h(b)\}$ to the lower and $\kappa \max_{[a,b]} h$ to the upper bound of the entropy sum, with $h(x) = -x \log_2 x$ evaluated by Arb and the maximum taken as in the entropy rule above. A block whose trace is certified to be below $t = 10^{-30}$ is not diagonalised: its entropy sum lies in $[0,\ t \log_2(d/t)]$. The program prints whether the ball for $\sum_k \operatorname{tr} \sigma_k$ contains 1; it did in every run, and the replay script of the release treats any other outcome as a failure.

What is trusted. The argument above; its implementation (Table IV); Arb's ball arithmetic, including its exponential, logarithm and square root; and the interpreter. The floating-point linear algebra of NumPy supplies hints only.

Appendix C: The second evaluator

This evaluator is not rigorous (Sec. IV D). In a frame with $p_X = p_Y$ put $f = p_X + p_Y$ and $c = p_I - p_Z$. With $s = x\bar{y}$, the image of $(|0\rangle + x|1\rangle)(\langle 0| + \bar{y}\langle 1|)$ is

$$Z = \begin{pmatrix} (1-f) + fs & c\,\bar{y} \\ c\,x & f + (1-f)s \end{pmatrix}, \qquad \det Z = f(1-f)(1+s^2) + \left[(1-f)^2 + f^2 - c^2\right] s, \tag{C1}$$

and $\sqrt{\binom{d}{\alpha}\binom{d}{\beta}}\,\mathrm{Sym}^d(Z)_{\alpha\beta}$ is the coefficient of $u^\alpha t^\beta$ in $(Z_{00} + Z_{01}t + Z_{10}u + Z_{11}ut)^d$. For $\alpha - \beta = \Delta \geq 0$ this coefficient is $x^\Delta P^{(d)}_{\alpha\beta}(s)$ with a polynomial P of degree at most $d - \Delta$ that obeys

$$P^{(d)}_{\alpha\beta} = Z_{00} P^{(d-1)}_{\alpha\beta} + cs\, P^{(d-1)}_{\alpha,\beta-1} + c\, P^{(d-1)}_{\alpha-1,\beta} + Z_{11} P^{(d-1)}_{\alpha-1,\beta-1}, \tag{C2}$$

where for $\Delta = 0$ the third term is $cs\, P^{(d-1)}_{\alpha,\alpha-1}$. All coefficients are nonnegative, since $(1-f)^2 - c^2 =$

TABLE IV. The certifying code. SHA-256 of the three files of the 128-bit evaluator (unchanged since their validation on 27 October 2026 and identical in versions 1 and 2 of the artifact), and the versions of the reference runs. The enclosures of Tables II and III were obtained with these versions; another FLINT version may change the last digits of an enclosure but not its validity.

`arb_ic.py`	`8a5887956e2b3814b500a3508815351aa6f7fa270d9fdbc8bd7b8f5e151fd5a1`
`rigcore.py`	`3a0142b4f95fbf2b8d4c5422ef7cf7e351c5d198f47cd84519c1f5357a7f3f7d`
`rigeig.py`	`4e84498eab2803d390eb150fab167e2177a40fc4fc433aae1dfc012fae56a286`
software	Python 3.14.3, python-flint 0.9.0 (FLINT 3.6.0), NumPy 2.5.4, mpmath 1.4.1, SciPy 1.18.1
verifier engine	Apple clang 21.0.0, `cc -O2 -ffp-contract=off -fno-fast-math`; arm64, macOS 26.5

$4p_I p_Z \geq 0$. The weighted block k of the image of $\sum_{ab} X_{ab}|D_a\rangle\langle D_b|$ is $m_k[\binom{d}{\alpha}\binom{d}{\beta}]^{-1/2}\sum_m P^{(d)}_{\alpha\beta,m}\,u^{(k,\Delta)}_m$ with $u^{(k,\Delta)}_m = \sum_b X_{b+\Delta,b}[\binom{n}{b+\Delta}\binom{n}{b}]^{-1/2}[s^{b-m}]\det(Z)^k$, and the latter obeys a three-term recurrence in k. The frame change uses the equivalent form $\langle D_b|W^{\otimes n}|D_a\rangle = 2^{-n/2}\sqrt{\binom{n}{b}/\binom{n}{a}}\,(-i)^{b-a}\sum_j(-1)^j\binom{n-b}{j}\binom{b}{a-j}$, derived separately and checked against $\exp(-i\pi J_x/2)$ computed from an eigendecomposition of J_x. Eigenvalues come from LAPACK (double precision) or from mpmath's Hermitian eigensolver (30 digits). The program imports NumPy and mpmath only.

Appendix D: All board records

[1] S. Lloyd, Capacity of the noisy quantum channel, Phys. Rev. A **55**, 1613 (1997).

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TABLE V. All 21 board records in order of posting. The verifier's I_c enclosure is in bits. Records #2–#28 were certified by version 1 of the verifier, #30 by version 2. Post numbers missing from the list are notes.

#	channel	n	p	I_c enclosure
2	depol.	150	0.06629494	$[1.5821, 2.1906]\text{e}{-9}$
3	X–Z	48	0.12179852	$[9.9207\text{e}{-9}, 1.0197\text{e}{-8}]$
4	2-Pauli	48	0.12086867	$[9.8282\text{e}{-9}, 1.0237\text{e}{-8}]$
6	depol.	240	0.06646039	$[2.9994, 4.4447]\text{e}{-9}$
9	X–Z	64	0.12331552	$[9.5442\text{e}{-9}, 1.0323\text{e}{-8}]$
10	2-Pauli	64	0.12235314	$[9.0366\text{e}{-9}, 1.0657\text{e}{-8}]$
11	depol.	240	0.06651279	$[1.0198\text{e}{-11}, 1.4597\text{e}{-9}]$
12	X–Z	64	0.12332757	$[2.4427\text{e}{-11}, 7.8861\text{e}{-10}]$
13	2-Pauli	64	0.12236279	$[1.3766\text{e}{-10}, 1.7557\text{e}{-9}]$
15	depol.	240	0.06662756	$[4.0998\text{e}{-12}, 2.1039\text{e}{-9}]$
16	2-Pauli	80	0.1233958	$[5.1234\text{e}{-11}, 7.1850\text{e}{-9}]$
17	X–Z	80	0.12437928	$[8.7420\text{e}{-12}, 2.4725\text{e}{-9}]$
18	depol.	240	0.06667253	$[9.7537\text{e}{-12}, 2.1687\text{e}{-9}]$
20	depol.	240	0.0666847	$[9.3916\text{e}{-12}, 2.1094\text{e}{-9}]$
21	2-Pauli	80	0.1236419	$[6.1335\text{e}{-11}, 7.6129\text{e}{-9}]$
22	X–Z	80	0.12459005	$[2.2419\text{e}{-11}, 2.6302\text{e}{-9}]$
24	X–Z	80	0.12463166	$[3.2608\text{e}{-12}, 2.6262\text{e}{-9}]$
25	2-Pauli	80	0.12367023	$[2.2868\text{e}{-12}, 7.5983\text{e}{-9}]$
27	2-Pauli	80	0.12367528	$[1.3223\text{e}{-13}, 7.5729\text{e}{-9}]$
28	X–Z	80	0.12464123	$[3.5713\text{e}{-11}, 2.6056\text{e}{-9}]$
30	depol.	480	0.06679861	$[2.4950\text{e}{-11}, 6.8873\text{e}{-9}]$

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